COMPUTER ORGANIZATION AND ARCHITECTURE (21CSS201T)

Unit 3 : Session 9 : SLO 1 — Restoring Division Algorithm (Procedure)
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Section A: Match the Following (Remember)

• A. Sign Bit → 4. Indicates positive or negative
• B. Negative Remainder → 5. Triggered when subtraction fails
• C. Quotient Register → 3. Stores result bits (0 or 1)
• D. Shift Step → 2. Logical left shift on registers
• E. Restoring Step → 1. Add divisor to restore remainder

Section B: Fill in the Blanks (Understand)

• The restoring division algorithm is primarily used for BINARY division of binary numbers.
• If subtraction in a cycle produces a NEGATIVE remainder, the algorithm restores the previous value.
• The REMAINDER register is shifted left with the quotient during each iteration.
• The divisor is SUBTRACTED from the remainder in each cycle.
• The quotient bit is set to 1 if the subtraction is successful.